

Direct Eight-Electron N_2O Electroreduction to NH_3 Enabled by an Fe Double-Atom Catalyst

Donghai Wu,[#] Kai Chen,[#] Peng Lv, Ziyu Ma, Ke Chu,^{*} and Dongwei Ma^{*}



Cite This: *Nano Lett.* 2024, 24, 8502–8509



Read Online

ACCESS |



Metrics & More



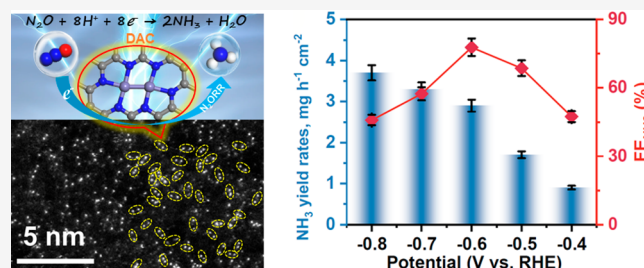
Article Recommendations



Supporting Information

ABSTRACT: N_2O is a dominant atmosphere pollutant, causing ozone depletion and global warming. Currently, electrochemical reduction of N_2O has gained increasing attention to remove N_2O , but its product is worthless N_2 . Here, we propose a direct eight-electron ($8e^-$) pathway to electrochemically convert N_2O into NH_3 . As a proof of concept, using density functional theory calculation, an Fe_2 double-atom catalyst (DAC) anchored by N-doped porous graphene ($\text{Fe}_2@\text{NG}$) was screened out to be the most active and selective catalyst for N_2O electroreduction toward NH_3 via the novel $8e^-$ pathway, which benefits from the unique bent N_2O adsorption configuration. Guided by theoretical prediction, $\text{Fe}_2@\text{NG}$ DAC was fabricated experimentally, and it can achieve a high N_2O -to- NH_3 Faradaic efficiency of 77.8% with a large NH_3 yield rate of $2.9 \text{ mg h}^{-1} \text{ cm}^{-2}$ at -0.6 V vs RHE in a neutral electrolyte. Our study offers a feasible strategy to synthesize NH_3 from pollutant N_2O with simultaneous N_2O removal.

KEYWORDS: N_2O electroreduction to NH_3 , double-atom catalysts, theoretical predictions, experimental validation



INTRODUCTION

Nitrous oxide (N_2O) has been recognized as the dominant ozone-depleting substance and meanwhile contributes significantly to global warming, due to the global warming potential of N_2O on a per mass basis being 310 times higher than that of CO_2 .^{1–3} Since the industrial revolution, the level of N_2O has increased gradually and significantly due to anthropogenic agriculture and industry activities.^{4–6} Therefore, it is of paramount importance to develop effective technology to eliminate N_2O .^{7–9} Currently, selective catalytic reduction is the most effective and widespread used technology for N_2O removal, which however requires valuable NH_3 and H_2 as reductants operating under high temperatures.^{10–12} It is hence imperative to conceptualize and develop new strategies for N_2O elimination.

Very recently, electrocatalytic N_2O reduction reaction (N_2ORR) under ambient conditions, utilizing renewable energies as a driving force and clean H_2O as reactant, has begun to gain attention for N_2O elimination. Especially, inspired by nitrous oxide reductase, N_2ORR has been investigated via the two-electron ($2e^-$) pathway: $\text{N}_2\text{O} + 2\text{H}^+ + 2e^- \rightarrow \text{N}_2 + \text{H}_2\text{O}$, $E^0 = 1.77 \text{ V}$ vs RHE.^{13–18} The electrocatalysts including transition-metal (TM) complexes, with atomically dispersed Re,¹⁵ Cu,¹⁷ and Fe¹⁸ as active centers and the bulk metal surface of AuPd alloys,¹⁶ Pd,¹⁹ and Ir,²⁰ were demonstrated to exhibit good electrocatalytic performance for $2e^- \text{N}_2\text{ORR}$. The high cost of noble metals and competing hydrogen evolution reaction (HER) however

impede their large-scale industrial applications. In addition, the product of $2e^- \text{N}_2\text{ORR}$, N_2 , has no use value.

Electrocatalytic NH_3 synthesis under ambient conditions has recently gained extensive research interest worldwide, as it can lower the reliance on fossil fuel and reduce CO_2 emission.^{21–23} So, we wonder, can N_2O be electrocatalytically reduced to more valuable NH_3 ? Herein, analogous to the electrochemical nitrate (NO_3^-) reduction reaction (NO_3RR : $\text{NO}_3^- + 9\text{H}^+ + 8e^- \rightarrow \text{NH}_3 + 3\text{H}_2\text{O}$) intensively studied recently,^{24–26} an eight-electron ($8e^-$) pathway toward NH_3 is proposed for N_2ORR : $\text{N}_2\text{O} + 8\text{H}^+ + 8e^- \rightarrow 2\text{NH}_3 + \text{H}_2\text{O}$.^{13,14} Based on the consumption of electrons per unit of NH_3 produced, $8e^- \text{N}_2\text{ORR}$ could have twice the market value compared with NO_3RR . Thus, we expect that developing efficient and low-cost electrocatalysts for $8e^- \text{N}_2\text{ORR}$ can not only eliminate harmful N_2O but also provide a green and effective method for NH_3 synthesis (Scheme 1). However, to the best of our knowledge, there is no comprehensive study on whether electrocatalytic reduction of N_2O toward NH_3 is feasible.

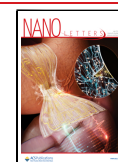
A double-atom catalyst (DAC) offers intrinsic advantages for multiple-step electrocatalytic reactions compared with the single-atom catalyst (SAC) counterpart.^{27–29} Both theoretical

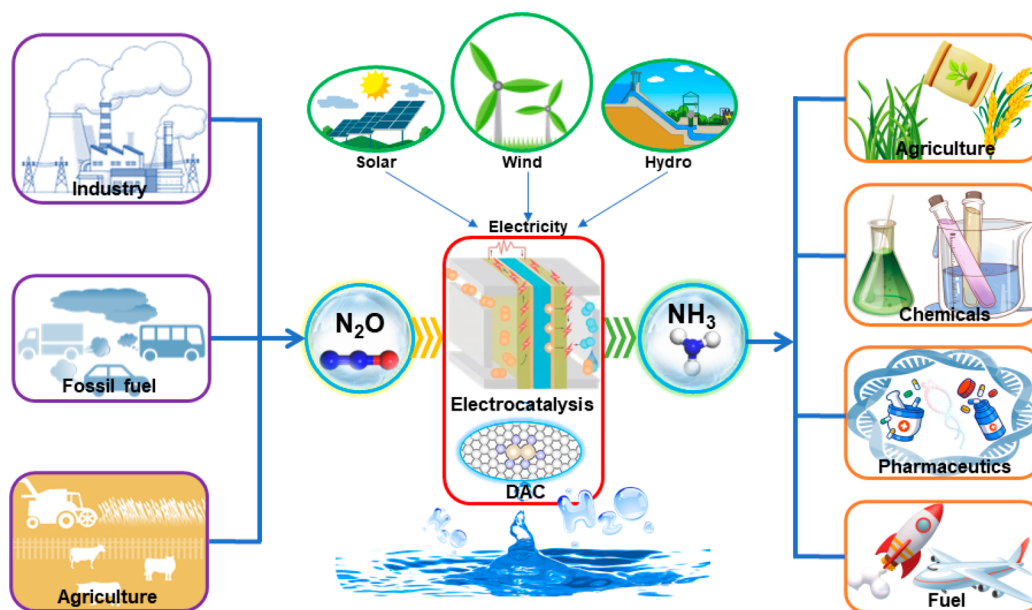
Received: February 1, 2024

Revised: June 26, 2024

Accepted: June 27, 2024

Published: July 1, 2024



Scheme 1. Schematic Illustration of Electrocatalytic N_2O Reduction toward NH_3 ^a

^a N_2O is supposed coming from exhausted gases in industrial factories and vehicles as well as agricultural production. The electricity can be supplied from renewable energies, such as solar, wind, and hydro power. The produced NH_3 can be used as the feedstock of fertilizers, chemicals, and pharmaceuticals as well as energy carriers.

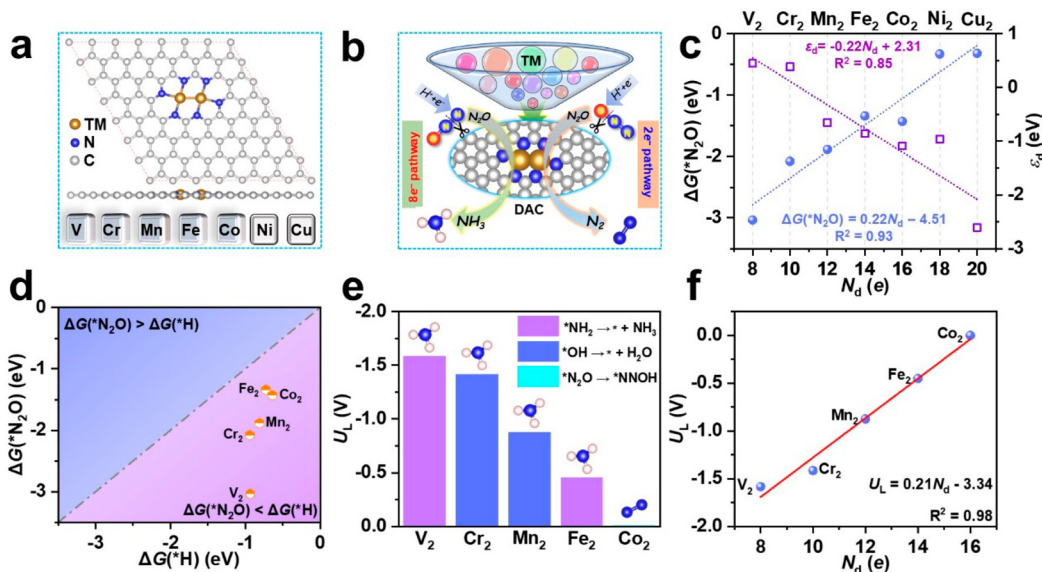


Figure 1. (a) Atomic model of the $\text{TM}_2\text{@NG}$ DACs. (b) Schematic illustration of the N_2ORR over DAC: through the $2e$ pathway toward N_2 or $8e$ pathway toward NH_3 . (c) The relationship between adsorption free energy of $^*\text{N}_2\text{O}$ ($\Delta G(^*\text{N}_2\text{O})$) as well as the d -band center (ϵ_d) and the number of valence electrons (N_d) of the TM_2 active center. (d) Comparison of the adsorption free energies of $^*\text{N}_2\text{O}$ with $^*\text{H}$ atom ($\Delta G(^*\text{H})$). (e) Summary of the limiting potential (U_L) and potential-determining step for N_2ORR on the DACs along the most favorable pathway. (f) Relationship between U_L and N_d .

and experimental studies have demonstrated that DACs can efficiently electrocatalytically convert CO_2 into valuable fuels or chemicals.^{30–32} Given that N_2O not only is isoelectronic with CO_2 but also has same linear configuration as CO_2 , one may expect that DACs could also achieve a satisfactory efficiency for N_2O electrocatalytic conversion. Herein, as a proof-of-concept study, we present a theory-guided rational design of graphene-based DACs for the electrocatalytic reduction of N_2O to efficiently produce NH_3 via the $8e$ pathway. Based on extensive density functional theory (DFT)

calculations, among a series of candidates, Fe_2 DACs anchored by the N-doped porous graphene (denoted as $\text{Fe}_2\text{@NG}$) were screened out, which were predicted to be highly active and selective toward NH_3 electrosynthesis. Guided by theoretical prediction, $\text{Fe}_2\text{@NG}$ DAC was experimentally synthesized by encapsulating an Fe dimer precursor within the cavity of ZIF and subsequent high-temperature pyrolysis, which stably delivers the N_2O -to- NH_3 Faradaic efficiency (FE_{NH_3}) of 77.8% and NH_3 yield rate of $2.9 \text{ mg h}^{-1} \text{ cm}^{-2}$ at -0.6 V vs RHE. This work proposes an alternative route for green and

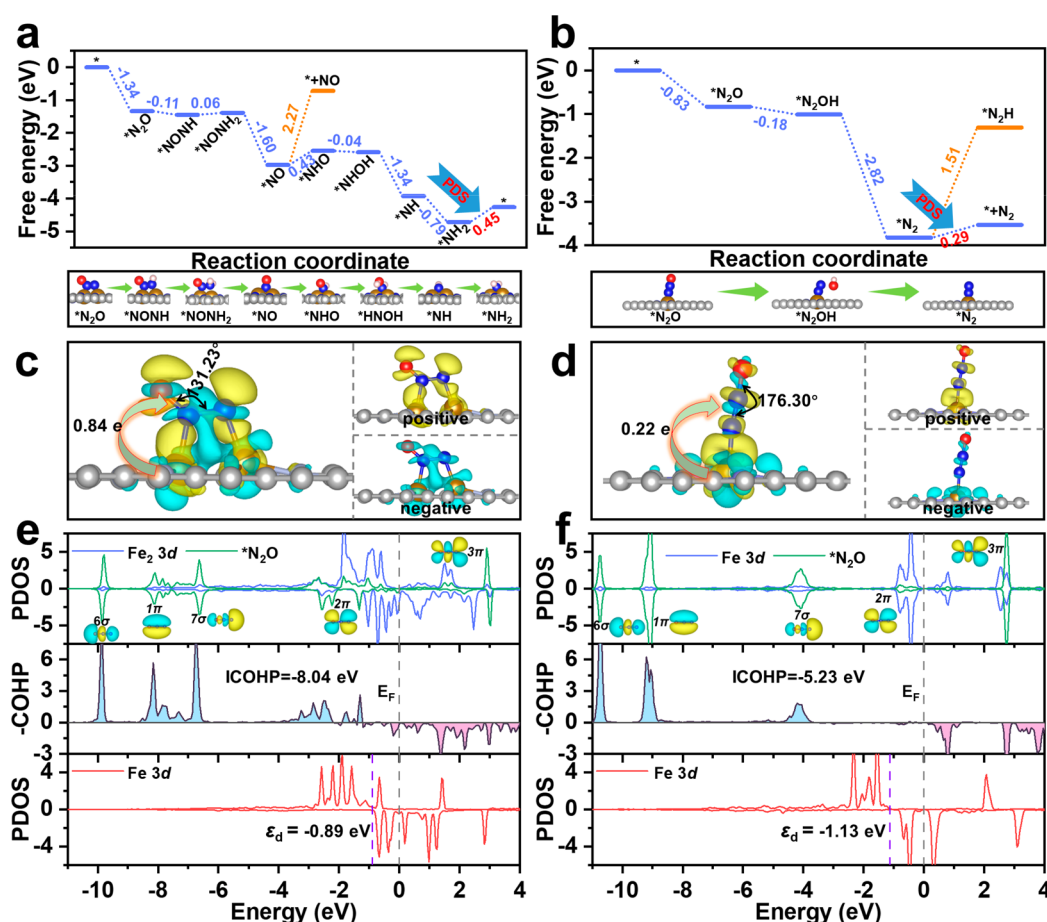


Figure 2. Free energy diagrams and key intermediate configurations for N₂ORR on Fe₂@NG DAC (a) and FeN₄ SAC (b). Charge density difference for *N₂O on Fe₂@NG DAC (c) and FeN₄ SAC (d), where the yellow and cyan regions denote electron accumulation and depletion, respectively. (e) Projected densities of states (PDOS, in states/eV) for the molecular orbitals of *N₂O and the 3d states of its bonded Fe atoms (top panel), the corresponding crystal orbital Hamilton population (COHP) (middle panel), and the PDOS of Fe 3d states of the pristine Fe₂@NG (bottom panel, the two Fe atoms are equivalent and only one is shown). Gray and purple vertical dashed lines represent Fermi level (E_F) and d -band center (ϵ_d), respectively. For COHP, blue and pink shadows denote bonding and antibonding interactions, respectively. (f) is similar to (e), except it is for FeN₄ SAC.

sustainable NH₃ synthesis from N₂O simultaneously along with N₂O removal.

First, we systematically investigated the experimental synthesis feasibility of TM₂@NG (TM including V, Cr, Mn, Fe, Co, Ni, and Cu) DACs based on DFT calculations, for which the atomic models are displayed in Figure 1a. Figure S1 shows that the values of formation energy (E_f) range from ~ 3.0 to 4.2 eV. Note that Mn₂@NG ($E_f = 3.65$ eV), Ni₂@NG ($E_f = 3.06$ eV),³⁰ and Cu₂@NG ($E_f = 3.78$ eV)³³ have been synthesized in experiment, and thus we anticipate that all the other DAC systems with comparable E_f can also be obtained. Further, the significant negative values (< -9.0 eV) of binding energy (Figure S2) suggest their high stabilities. In addition, the key parameters of these systems have been compiled in Table S1, and their total electronic densities of states (DOS) are displayed in Figure S3. The metallic state characteristics are beneficial for charge transfer during N₂ORR.³⁴ Consequently, we will screen the promising candidates from all the proposed DACs for the efficient electrocatalytic conversion of N₂O into NH₃ (Figure 1b).

Effective adsorption and activation of N₂O is a prerequisite for efficient N₂ORR. For this, various initial configurations for N₂O on the DACs have been considered, as shown in Figure S4, and the most stable adsorption configurations for *N₂O (*

denotes the adsorption site) are shown in Figure S5. *N₂O energetically prefers to adopt the side-on adsorption pattern on V₂, Cr₂, Mn₂, and Fe₂@NG DACs by forming multiple chemical bonds, while it adopts the end-on adsorption pattern on Co₂@NG with two N–Co bonds. However, for Ni₂ and Cu₂@NG DACs, the N₂O molecules are physisorbed, as suggested by large distances between molecules and substrates. The calculated adsorption free energies for *N₂O ($\Delta G(*N_2O)$) are presented in Figure 1c. We can observe a good linear scaling relationship between $\Delta G(*N_2O)$ and the number of valence electrons (N_d) of the TM₂ active center from V₂ to Cu₂@NG. The fewer valence electrons the TM₂ active center has, the stronger the *N₂O binding becomes, which can be ascribed to the d -band center (ϵ_d) of the TM₂ active center shifting up with respect to the Fermi level from Cu₂ to V₂@NG (Figure 1c).³⁵ HER is the main competing reaction limiting the efficiency of the electrocatalytic N₂ORR process. Similar to NO₃RR,^{36,37} the thermodynamic selectivity trend between HER and N₂ORR can be evaluated by comparing the binding strengths of *H and *N₂O. The results presented in Figure 1d strongly indicate that for all the considered DAC systems HER could be well hindered and N₂ORR should be dominant, due to *N₂O binding more strongly than *H.

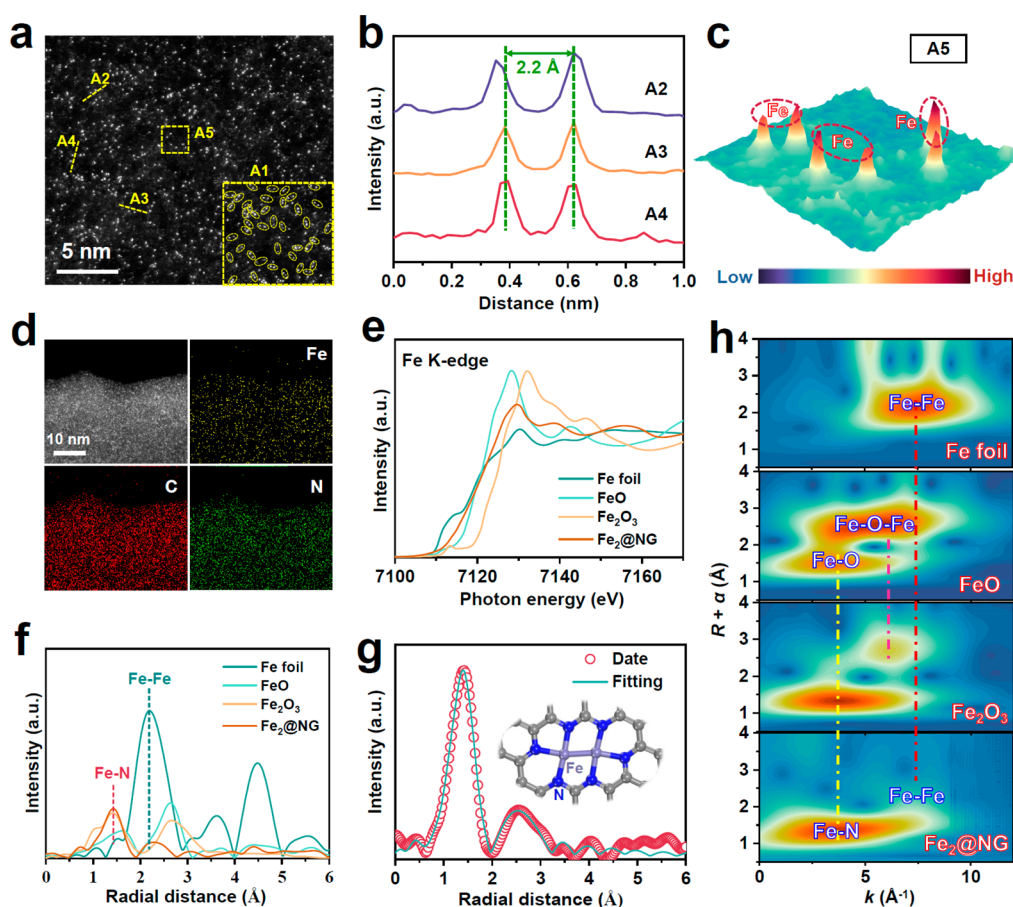


Figure 3. Characterizations of Fe₂@NG DAC: (a) AC-HAADF-STEM image and corresponding (b) atomic intensity line scanning analysis and (c) 3D topographic atom image. (d) Element mapping images. (e) Fe K-edge XANES, (f) EXAFS spectra, and (h) WT profiles of Fe₂@NG and reference samples. (g) EXAFS fitting result with the fitting model inset.

We further fully explored the detailed reaction mechanism of N₂ORR on V₂, Cr₂, Mn₂, Fe₂, and Co₂@NG DACs, for which various hydrogenation sites of the intermediates have been considered, and the most stable configurations were selected to study the subsequent elementary reaction. Note that Ni₂ and Cu₂@NG DACs have not been considered because they are inert toward N₂O adsorption. The calculated limiting potential (U_L) and the corresponding potential-determining step (PDS) are summarized in Figure 1e. Interestingly, for all the systems, valuable NH₃ is the reaction product, except for Co₂@NG DAC, which generates N₂. Especially, the final product is closely related to the adsorption pattern of *N₂O. DACs with *N₂O side-on adsorption produce valuable NH₃ through the 8e N₂ORR pathway, while that with an end-on pattern generates worthless N₂ via a 2e N₂ORR pathway. This is probably due to the fact that for the end-on configuration the N–O bond in *N₂O can be easily cleaved upon hydrogenation to leave *N₂ desorption, while for the side-on configuration, the N–N bond in *N₂O is first cleaved during the protonation process to form NH₃, as schematically shown in Figure 1b. From Figure 1e, for V₂ and Fe₂@NG DACs, *NH₂ → * + NH₃ is identified as PDS; for Cr₂ and Mn₂@NG DACs, *OH → * + H₂O is identified as PDS; for Co₂@NG DAC, *N₂O → *NNOH is PDS. Significantly, among all the studied systems, Fe₂@NG DAC was screened out to be the most promising candidate to perform 8e N₂ORR producing NH₃, which exhibits a rather low U_L = −0.45 V. Moreover, from Figure S6, there is a linear correlation between U_L and $\Delta G(*N_2O)$.

Considering that $\Delta G(*N_2O)$ linearly scales with N_d of the TM₂ active center (Figure 1c), N_d can be used as the activity descriptor for N₂ORR on the DACs, as shown in Figure 1f. Besides N_d , the *d*-band center, and polarized charge (ΔQ) of the TM₂ active center can also serve as the activity descriptor for N₂ORR (see Figures S7 and S8).

To get insight into the reaction mechanism of N₂ORR on various DACs, we carefully examined the reaction free energy diagram. The results for Fe₂@NG are presented in Figure 2a, and those for other systems in Figures S9–S12. N₂O can be strongly adsorbed on Fe₂@NG (Figure 2a) with a downhill free energy of −1.34 eV, suggesting that N₂O can be effectively captured and activated by the dual Fe active center. For the first hydrogenation step of *N₂O, the proton–electron pair prefers to attack the N atom away from the O atom in *N₂O with a downhill free energy of −0.11 eV. In the second hydrogenation step, the proton–electron pair continues to attack the same N atom in *N₂O to form *NONH₂. Subsequently, the formed *NONH₂ species is hydrogenated to first produce an NH₃ molecule, leaving *NO on Fe₂@NG, with a downhill free energy of −1.60 eV. In the following, the reaction proceeds along the well-studied NO reduction reaction pathway.^{38,39} *NO is hydrogenated to *NHO with an uphill free energy of 0.43 eV. Then two proton–electron pairs continuously attack the O atom to form a H₂O molecule and *NH species with a downhill free energy of −1.34 eV. Consequently, the last hydrogenation step is the PDS for 8e N₂ORR on Fe₂@NG with a rather small U_L of −0.45 V. In

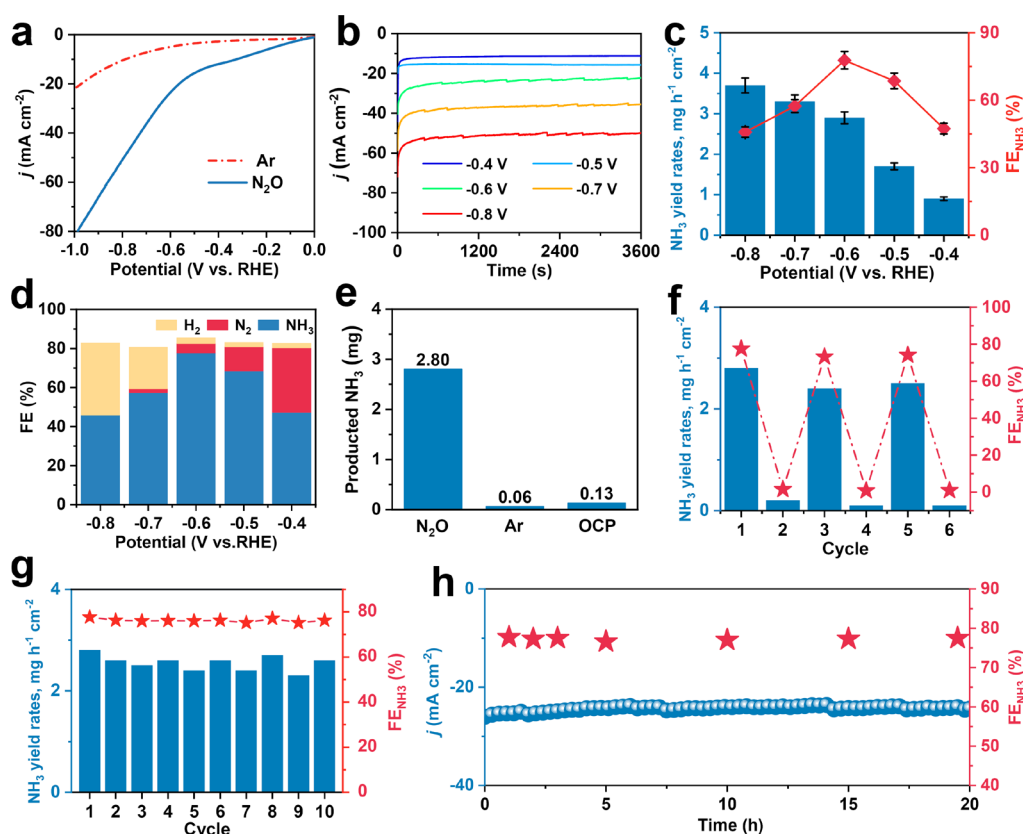


Figure 4. (a) LSV curves of Fe₂@NG DAC in Ar/N₂O-saturated 0.5 M Na₂SO₄. Chronoamperometry test at various potentials (b) and obtained NH₃ yield rates and FE_{NH3} (c) on Fe₂@NG. (d) FEs of various products. (e) Amounts of produced NH₃ under different conditions. (f) N₂O–Ar switching experiment, (g) cycling, and (h) long-term chronoamperometry test of Fe₂@NG at –0.6 V.

addition, due to the excessive adsorption of *NO with an adsorption free energy of –2.27 eV, the formation of byproduct NO can be excluded. As a comparison, we further studied the N₂ORR reaction mechanism on the FeN₄ SAC (Fe atom is coordinated with four pyridinic N atoms). The free energy diagram in Figure 2b indicates that *N₂O binds relatively weakly on FeN₄ SAC compared with Fe₂@NG and prefers to be reduced into N₂ via the 2e pathway, due to the *N₂ hydrogenation being much more difficult than its desorption by 1.22 eV.

The distinct N₂ORR reaction mechanism on the Fe₂@NG DAC and FeN₄ SAC was revealed by electronic structure analysis. The effective activation of N₂O is a crucial step for its further hydrogenation. Although N₂O is an exceedingly poor ligand, as shown in Figure 2c, *N₂O on Fe₂@NG adopts a rarely observed bent configuration with an O–N–N bond angle of 131.23° (identical to the η²-N,N'-coordinated N₂O in ref 40), thanks to the neighboring dual active site. For this configuration, the N₂O molecule can be largely activated, which can be inferred from the charge redistribution upon its adsorption, i.e., significant electron accumulation (yellow region) on the antibonding orbital of *N₂O and electron depletion (cyan region) on its bonding orbital, suggesting that N₂O adsorption follows the charge “acceptance–donation” mechanism. Consequently, the N–N bond in *N₂O is significantly elongated to 1.26 Å from 1.13 Å in its free state, which should be responsible for its unique hydrogenation pathway. Conversely, with the same activation mechanism, *N₂O on FeN₄ SAC adopts an end-on adsorption configuration and can be only weakly activated compared with Fe₂@

NG DAC. The *N₂O almost keeps its linear configuration, and its N–N bond is only slightly increased by 0.02 Å. Moreover, the charge transfer for FeN₄ resulting from *N₂O is four times less than that for Fe₂@NG (0.22 vs 0.84 e). This end-on adsorption configuration is conducive to the hydrogenation of the terminal O atom in *N₂O, agreeing with previous studies.^{17,18} The above picture can be further supported by the DOS and crystal orbital Hamiltonian population (COHP) analysis (Figure 2e and 2f).⁴¹ More negative integrated COHP (ICOHP) for *N₂O on Fe₂@NG (–8.04 eV) compared with that on FeN₄ (–5.23 eV) is mainly responsible for the binding strength difference. Especially, compared with FeN₄ SAC, the much greater occupation of an antibonding state near the Fermi level (pink shadow region) for Fe₂@NG DAC further suggests the higher activation level of *N₂O. Finally, from the bottom panel of Figure 2e and 2f, we can see the d-band center of the Fe₂ active center in Fe₂@NG (–0.89 eV) is much higher than that of the Fe active center in FeN₄ (–1.13 eV). Thus, it can be concluded that the favorable geometric and electronic structures of dual Fe sites promote the 8e N₂ORR pathway to produce NH₃.

Inspired by the above DFT investigations, we further proceeded with the proof-of-concept experiments. Fe₂@NG DAC (Fe: 1.77 at. %) was synthesized by encapsulating the Fe dimer precursor within the cavity of ZIF, followed by high-temperature pyrolysis to yield Fe₂@NG.³³ We employed spherical aberration-corrected high-angle annular dark-field scanning transmission electron microscopy (AC-HAADF-STEM) to analyze the atomic arrangement of Fe species in Fe₂@NG. It can be seen from Figure 3a that numerous

atomically dispersed bright dots, attributed to isolated Fe atoms, are evenly anchored onto the carbon matrix. No evident Fe clusters can be observed. Besides, by selecting the A1 region in Figure 3a as a representative example, a large proportion of two neighboring Fe atoms as diatomic Fe sites (highlighted by yellow circles) can be found. Meanwhile, the atomic intensity profiles (A2–4 lines in Figure 3b) along with the 3D topographic atom image (A5 region in Figure 3c) provided a clearer depiction of diatomic Fe sites in Fe₂@NG. The average distance between adjacent Fe atoms of the diatomic Fe sites is measured to be approximately 2.20 Å, correlating well with the DFT model (2.22 Å) and previous reports.^{33,42} The elemental mapping images of Fe₂@NG (Figure 3d) presented a uniform distribution of Fe species on the carbon substrate.

Then the electronic structures and coordination environment of Fe₂@NG DAC were investigated using X-ray absorption spectroscopy (XAS) measurements. The Fe K-edge XANES spectra revealed that the absorption edge of Fe₂@NG is situated between that of Fe foil and FeO reference (Figure 3e), indicating that the oxidation state of Fe in Fe₂@NG falls within the range from 0 to +2. In the EXAFS spectrum of Fe₂@NG, the primary peak appearing at 1.43 Å (Figure 3f) corresponds to Fe–N coordination. Additionally, a second-shell peak assigned to the Fe–Fe bond was observed at 2.25 Å, which can be attributed to the interaction between diatomic Fe sites, consistent with the AC-HAADF-STEM results.⁴³ Likewise, the corresponding wavelet transform (WT) contour maps (Figure 3h) show the prominent Fe–N intensity maximum and slight Fe–Fe intensity maximum in Fe₂@NG, thus confirming, once again, the existence of both Fe–N and Fe–Fe coordination bonds in Fe₂@NG. The EXAFS fitting analysis (Figure 3g and Table S2) revealed that the coordination numbers are ~4 for Fe–N coordination and ~1 for Fe–Fe coordination, indicating that the Fe₂@NG comprises two adjacent Fe₁–N₃ units, where each isolated Fe atom of diatomic Fe sites is coordinated with three surrounding N atoms and diatomic Fe bonds with each other.

The N₂ORR performance of Fe₂@NG DAC was assessed in a gastight H-cell containing 0.5 M Na₂SO₄ (Figure S13).⁴⁴ Linear sweep voltammetry (LSV, Figure 4a) curves of Fe₂@NG exhibited a noticeable enhancement in current density (*j*) in N₂O-saturated electrolyte compared to Ar-saturated electrolyte, suggesting the high catalytic activity of Fe₂@NG toward the N₂ORR.³⁸ Chronoamperometry tests (Figure 4b, 1 h electrolysis) were conducted to quantitatively evaluate the N₂ORR performance of Fe₂@NG. Liquid (NH₃) and gas (N₂, H₂) products generated during the electrolysis were detected using the colorimetric method (Figure S14) and gas chromatography, respectively. As depicted in Figure 4c, Fe₂@NG achieved a maximum FE_{NH₃} of 77.8% at –0.6 V with a corresponding NH₃ yield rate of 2.9 mg h^{–1} cm^{–2}. Notably, at lower potentials, N₂ serves as the primary byproduct (Figure 4d). When the potential reaches –0.6 V, FE_{N₂} becomes negligible, underscoring the high selectivity of Fe₂@NG for direct 8e N₂O-to-NH₃ conversion. H₂ becomes the predominant byproduct beyond –0.6 V, attributed to the enhanced HER at higher potentials. Similar trends are also evident in the results of partial current density (Figure S15). For comparison, a single-atom Fe catalyst (Figure S16) with isolated Fe₁–N₄ sites was synthesized and its N₂ORR performances were evaluated. At the potential range from –0.4 to –0.8 V, FeN₄ SAC exhibited a dominance of FE_{N₂} and considerably lower FE_{NH₃} (Figure S17). These results highlight

the great advantage of Fe₂@NG DAC in catalyzing the 8e N₂ORR for NH₃ synthesis.

Several control tests were carried out to determine the source of NH₃ during the N₂ORR electrolysis. As shown in Figure 4e, electrolysis measurements conducted under Ar conditions or at open-circuit potential (OCP) yielded negligible amounts of NH₃. The N₂O–Ar switching test (Figure 4f) highlighted the substantial NH₃ production during the N₂O cycles, while NH₃ was nearly undetectable in the Ar cycles. These findings provide strong evidence that the generated NH₃ unequivocally originates from the N₂ORR process catalyzed by Fe₂@NG DAC. We also performed the electrochemical measurements of Fe₂@NG in a flow cell (Figure S18), in which the primary product on Fe₂@NG remains NH₃ rather than N₂ due to the thermodynamic favorability of NH₃ synthesis over N₂ synthesis on Fe₂@NG.

In addition to its excellent electrocatalytic activity and selectivity, Fe₂@NG DAC also exhibits an exceptional durability throughout the N₂ORR process. As presented in the 10 consecutive electrolysis cycles (Figure 4g), no significant fluctuations in FE_{NH₃} and NH₃ yield rates were observed, affirming the favorable cycling stability of Fe₂@NG. The extended-duration chronoamperometry test (Figure 4h) revealed a minimal decay in *j* and slight fluctuations in FE_{NH₃} over 20 h of continuous electrolysis, underscoring the excellent long-term stability of Fe₂@NG DAC. These experimental results validate the theoretical predictions and demonstrate the great potential of Fe₂@NG DAC as an active, selective, and durable N₂ORR catalyst to promote N₂O-to-NH₃ electrosynthesis.

CONCLUSIONS

In summary, we have theoretically proposed and experimentally verified a new 8e N₂ORR pathway to electrocatalytically convert N₂O into valuable NH₃ with N₂O removed simultaneously. To be specific, Fe₂@NG DAC was screened out via extensive DFT calculations as the best one among a series of candidates. The underlying reason can be ascribed to the unique dual-atom active site with suitable electronic structure of the anchored Fe₂ motif, which makes N₂O adsorption adopt a favorable bent configuration, beneficial for the subsequent N₂O hydrogenation reaction via the 8e pathway to produce NH₃. Then, we successfully fabricated Fe₂@NG DAC, which delivers a maximal FE of 77.8% and a large NH₃ yield of 2.9 mg h^{–1} cm^{–2} at the potential of –0.6 V vs RHE in a neutral electrolyte. Our work offers an alternative electrocatalytic route for green and sustainable NH₃ synthesis from the pollutant N₂O, which will in the meantime mitigate the N₂O level in the atmosphere.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.4c00576>.

Experimental and calculation details, additional characterizations of Fe₂@NG DAC, additional electrochemical measurements, additional DFT computations (PDF)

AUTHOR INFORMATION

Corresponding Authors

Ke Chu – School of Materials Science and Engineering, Lanzhou Jiaotong University, Lanzhou 730070, China;

orcid.org/0000-0002-0606-2689; Email: chuk630@mail.lzjtu.cn

Dongwei Ma – Anhui Province Industrial Generic Technology Research Center for Alumics Materials, School of Physics and Electronic Information, Huaibei Normal University, Huaibei, Anhui 235000, China; Key Laboratory for Special Functional Materials of Ministry of Education, and School of Materials Science and Engineering, Henan University, Kaifeng 475004, China; Email: dwmachina@126.com

Authors

Donghai Wu – Anhui Province Industrial Generic Technology Research Center for Alumics Materials, School of Physics and Electronic Information, Huaibei Normal University, Huaibei, Anhui 235000, China; Key Laboratory for Special Functional Materials of Ministry of Education, and School of Materials Science and Engineering, Henan University, Kaifeng 475004, China

Kai Chen – School of Materials Science and Engineering, Lanzhou Jiaotong University, Lanzhou 730070, China

Peng Lv – Key Laboratory for Special Functional Materials of Ministry of Education, and School of Materials Science and Engineering, Henan University, Kaifeng 475004, China

Ziyu Ma – Key Laboratory for Special Functional Materials of Ministry of Education, and School of Materials Science and Engineering, Henan University, Kaifeng 475004, China

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.nanolett.4c00576>

Author Contributions

[#]D.W. and K.C. contributed equally to this work.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work is supported by the National Natural Science Foundation of China (Grant 22102065, 52161025) and the Program for Science & Technology Innovation Talents in Universities of Henan Province (Grant No. 20HASTIT028).

REFERENCES

- (1) Ravishankara, A. R.; Daniel, J. S.; Portmann, R. W. Nitrous Oxide (N₂O): The Dominant Ozone-Depleting Substance Emitted in the 21st Century. *Science* **2009**, *326*, 123–125.
- (2) Montzka, S. A.; Dlugokencky, E. J.; Butler, J. H. Non-CO₂ greenhouse gases and climate change. *Nature* **2011**, *476*, 43–50.
- (3) Davidson, E. A.; Winiwarter, W. Urgent abatement of industrial sources of nitrous oxide. *Nat. Clim. Chang.* **2023**, *13*, 599–601.
- (4) Thompson, R. L.; Lassaletta, L.; Patra, P. K.; Wilson, C.; Wells, K. C.; Gressent, A.; Koffi, E. N.; Chipperfield, M. P.; Winiwarter, W.; Davidson, E. A.; Tian, H.; Canadell, J. G. Acceleration of global N₂O emissions seen from two decades of atmospheric inversion. *Nat. Clim. Chang.* **2019**, *9*, 993–998.
- (5) Tian, H.; Xu, R.; Canadell, J. G.; Thompson, R. L.; Winiwarter, W.; Suntharalingam, P.; Davidson, E. A.; Ciais, P.; Jackson, R. B.; Janssens-Maenhout, G.; Prather, M. J.; Regnier, P.; Pan, N.; Pan, S.; Peters, G. P.; Shi, H.; Tubiello, F. N.; Zaehle, S.; Zhou, F.; Arneeth, A.; Battaglia, G.; Berthet, S.; Bopp, L.; Bouwman, A. F.; Buitenhuis, E. T.; Chang, J.; Chipperfield, M. P.; Dangal, S. R. S.; Dlugokencky, E.; Elkins, J. W.; Eyre, B. D.; Fu, B.; Hall, B.; Ito, A.; Joos, F.; Krummel, P. B.; Landolfi, A.; Laruelle, G. G.; Lauerwald, R.; Li, W.; Lienert, S.; Maavara, T.; MacLeod, M.; Millet, D. B.; Olin, S.; Patra, P. K.; Prinn, R. G.; Raymond, P. A.; Ruiz, D. J.; van der Werf, G. R.; Vuichard, N.; Wang, J.; Weiss, R. F.; Wells, K. C.; Wilson, C.; Yang, J.; Yao, Y. A

comprehensive quantification of global nitrous oxide sources and sinks. *Nature* **2020**, *586*, 248–256.

(6) Bols, M. L.; Snyder, B. E. R.; Rhoda, H. M.; Cnudde, P.; Fayad, G.; Schoonheydt, R. A.; Van Speybroeck, V.; Solomon, E. I.; Sels, B. F. Coordination and activation of nitrous oxide by iron zeolites. *Nat. Catal.* **2021**, *4*, 332–340.

(7) Alves, L.; Holz, L. I. V.; Fernandes, C.; Ribeirinha, P.; Mendes, D.; Fagg, D. P.; Mendes, A. A comprehensive review of NO_x and N₂O mitigation from industrial streams. *Renew. Sust. Energy Rev.* **2022**, *155*, No. 111916.

(8) Le Vaillant, F.; Mateos Calbet, A.; González-Pelayo, S.; Reijerse, E. J.; Ni, S.; Busch, J.; Cornella, J. Catalytic synthesis of phenols with nitrous oxide. *Nature* **2022**, *604*, 677–683.

(9) Kieger, S.; Delahay, G.; Coq, B.; Neveu, B. Selective Catalytic Reduction of Nitric Oxide by Ammonia over Cu-FAU Catalysts in Oxygen-Rich Atmosphere. *J. Catal.* **1999**, *183*, 267–280.

(10) Zeng, R.; Feller, M.; Diskin-Posner, Y.; Shimon, L. J. W.; Ben-David, Y.; Milstein, D. CO Oxidation by N₂O Homogeneously Catalyzed by Ruthenium Hydride Pincer Complexes Indicating a New Mechanism. *J. Am. Chem. Soc.* **2018**, *140*, 7061–7064.

(11) Zhang, X.; Shen, Q.; He, C.; Ma, C.; Cheng, J.; Li, L.; Hao, Z. Investigation of Selective Catalytic Reduction of N₂O by NH₃ over an Fe-Mordenite Catalyst: Reaction Mechanism and O₂ Effect. *ACS Catal.* **2012**, *2*, 512–520.

(12) Park, Y.-K.; Kim, B.-S. Catalytic removal of nitrogen oxides (NO, NO₂, N₂O) from ammonia-fueled combustion exhaust: A review of applicable technologies. *Chem. Eng. J.* **2023**, *461*, No. 141958.

(13) Bratsch, S. G. Standard Electrode Potentials and Temperature Coefficients in Water at 298.15 K. *J. Phys. Chem. Ref. Data* **1989**, *18*, 1–21.

(14) Ko, B. H.; Hasa, B.; Shin, H.; Zhao, Y.; Jiao, F. Electrochemical Reduction of Gaseous Nitrogen Oxides on Transition Metals at Ambient Conditions. *J. Am. Chem. Soc.* **2022**, *144*, 1258–1266.

(15) Deeba, R.; Molton, F.; Chardon-Noblat, S.; Costentin, C. Effective Homogeneous Catalysis of Electrochemical Reduction of Nitrous Oxide to Dinitrogen at Rhodium Carbonyl Catalysts. *ACS Catal.* **2021**, *11*, 6099–6103.

(16) Kim, K.; Byun, J.; Kim, H.; Lee, K.-S.; Lee, H. S.; Kim, J.; Hyeon, T.; Kim, J. J.; Han, J. W. Systematic Approach to Designing a Highly Efficient Core-Shell Electrocatalyst for N₂O Reduction. *ACS Catal.* **2021**, *11*, 15089–15097.

(17) Martinez, J. L.; Schneider, J. E.; Anferov, S. W.; Anderson, J. S. Electrochemical Reduction of N₂O with a Molecular Copper Catalyst. *ACS Catal.* **2023**, *13*, 12673–12680.

(18) Stanley, J. S.; Wang, X. S.; Yang, J. Y. Selective Electrocatalytic Reduction of Nitrous Oxide to Dinitrogen with an Iron Porphyrin Complex. *ACS Catal.* **2023**, *13*, 12617–12622.

(19) Kudo, A.; Mine, A. Electrocatalytic reduction of nitrous oxide on metal and oxide electrodes in aqueous solution. *Appl. Surf. Sci.* **1997**, *121*, 538–542.

(20) Gómez, R.; Weaver, M. J. Reduction of Nitrous Oxide on Iridium Single-Crystal Electrodes. *Langmuir* **2002**, *18*, 4426–4432.

(21) Liang, J.; Liu, Q.; Alshehri, A. A.; Sun, X. Recent advances in nanostructured heterogeneous catalysts for N-cycle electrocatalysis. *Nano Res. Energy* **2022**, *1*, No. 9120010.

(22) Wang, Y.; Wang, C.; Li, M.; Yu, Y.; Zhang, B. Nitrate electroreduction: mechanism insight, in situ characterization, performance evaluation, and challenges. *Chem. Soc. Rev.* **2021**, *50*, 6720–6733.

(23) Zhang, G.; Li, X.; Chen, K.; Guo, Y.; Ma, D.; Chu, K. Tandem Electrocatalytic Nitrate Reduction to Ammonia on MBenes. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202300054.

(24) Chen, F.-Y.; Wu, Z.-Y.; Gupta, S.; Rivera, D. J.; Lambeets, S. V.; Pecaut, S.; Kim, J. Y. T.; Zhu, P.; Finfrook, Y. Z.; Meira, D. M.; King, G.; Gao, G.; Xu, W.; Cullen, D. A.; Zhou, H.; Han, Y.; Perea, D. E.; Muhich, C. L.; Wang, H. Efficient conversion of low-concentration nitrate sources into ammonia on a Ru-dispersed Cu nanowire electrocatalyst. *Nat. Nanotechnol.* **2022**, *17*, 759–767.

- (25) Li, X.; Shen, P.; Li, X.; Ma, D.; Chu, K. Sub-nm RuO_x Clusters on Pd Metallene for Synergistically Enhanced Nitrate Electroreduction to Ammonia. *ACS Nano* **2023**, *17*, 1081–1090.
- (26) Du, H.; Guo, H.; Wang, K.; Du, X.; Beshiwork, B. A.; Sun, S.; Luo, Y.; Liu, Q.; Li, T.; Sun, X. Durable Electrocatalytic Reduction of Nitrate to Ammonia over Defective Pseudobrookite Fe₂TiO₅ Nanofibers with Abundant Oxygen Vacancies. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202215782.
- (27) Wu, D.; He, B.; Wang, Y.; Lv, P.; Ma, D.; Jia, Y. Double-atom catalysts for energy-related electrocatalysis applications: a theoretical perspective. *J. Phys. D Appl. Phys.* **2022**, *55*, No. 203001.
- (28) Ying, Y.; Luo, X.; Qiao, J.; Huang, H. More is Different: Synergistic Effect and Structural Engineering in Double-Atom Catalysts. *Adv. Funct. Mater.* **2021**, *31*, No. 2007423.
- (29) Li, L.; Yuan, K.; Chen, Y. Breaking the Scaling Relationship Limit: From Single-Atom to Dual-Atom Catalysts. *Acc. Mater. Res.* **2022**, *3*, 584–596.
- (30) Hao, Q.; Zhong, H.-x.; Wang, J.-z.; Liu, K.-h.; Yan, J.-m.; Ren, Z.-h.; Zhou, N.; Zhao, X.; Zhang, H.; Liu, D.-x.; Liu, X.; Chen, L.-w.; Luo, J.; Zhang, X.-b. Nickel dual-atom sites for electrochemical carbon dioxide reduction. *Nat. Synth.* **2022**, *1*, 719–728.
- (31) Li, H.; Wu, D.; Wu, J.; Song, Y.; Lv, W.; Duan, Z.; Ma, D. Mechanistic understanding of the electrocatalytic conversion of CO into C₂₊ products by double-atom catalysts. *Mater. Today Phys.* **2023**, *37*, No. 101203.
- (32) Fan, Z.; Luo, R.; Zhang, Y.; Zhang, B.; Zhai, P.; Zhang, Y.; Wang, C.; Gao, J.; Zhou, W.; Sun, L.; Hou, J. Oxygen-Bridged Indium-Nickel Atomic Pair as Dual-Metal Active Sites Enabling Synergistic Electrocatalytic CO₂ Reduction. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202216326.
- (33) Leng, K.; Zhang, J.; Wang, Y.; Li, D.; Bai, L.; Shi, J.; Li, X.; Zheng, L.; Bai, J.; Qu, Y. Interfacial Cladding Engineering Suppresses Atomic Thermal Migration to Fabricate Well-Defined Dual-Atom Electrocatalysts. *Adv. Funct. Mater.* **2022**, *32*, No. 2205637.
- (34) Guo, X.; Gu, J.; Lin, S.; Zhang, S.; Chen, Z.; Huang, S. Tackling the Activity and Selectivity Challenges of Electrocatalysts toward the Nitrogen Reduction Reaction via Atomically Dispersed Biatom Catalysts. *J. Am. Chem. Soc.* **2020**, *142*, 5709–5721.
- (35) Hammer, B.; Morikawa, Y.; Nørskov, J. K. CO Chemisorption at Metal Surfaces and Overlayers. *Phys. Rev. Lett.* **1996**, *76*, 2141–2144.
- (36) Niu, H.; Zhang, Z.; Wang, X.; Wan, X.; Shao, C.; Guo, Y. Theoretical Insights into the Mechanism of Selective Nitrate-to-Ammonia Electroreduction on Single-Atom Catalysts. *Adv. Funct. Mater.* **2021**, *31*, No. 2008533.
- (37) Wang, Y.; Zhou, W.; Jia, R.; Yu, Y.; Zhang, B. Unveiling the Activity Origin of a Copper-based Electrocatalyst for Selective Nitrate Reduction to Ammonia. *Angew. Chem., Int. Ed.* **2020**, *59*, 5350–5354.
- (38) Chen, K.; Zhang, Y.; Xiang, J.; Zhao, X.; Li, X.; Chu, K. p-Block Antimony Single-Atom Catalysts for Nitric Oxide Electroreduction to Ammonia. *ACS Energy Lett.* **2023**, *8*, 1281–1288.
- (39) Zhang, L.; Liang, J.; Wang, Y.; Mou, T.; Lin, Y.; Yue, L.; Li, T.; Liu, Q.; Luo, Y.; Li, N.; Tang, B.; Liu, Y.; Gao, S.; Alshehri, A. A.; Guo, X.; Ma, D.; Sun, X. High-Performance Electrochemical NO Reduction into NH₃ by MoS₂ Nanosheet. *Angew. Chem., Int. Ed.* **2021**, *60*, 25263–25268.
- (40) Puerta Lombardi, B. M.; Gendy, C.; Gelfand, B. S.; Bernard, G. M.; Wasylshen, R. E.; Tuononen, H. M.; Roesler, R. Side-on Coordination in Isostructural Nitrous Oxide and Carbon Dioxide Complexes of Nickel. *Angew. Chem., Int. Ed.* **2021**, *60*, 7077–7081.
- (41) Deringer, V. L.; Tchougréeff, A. L.; Dronskowski, R. Crystal orbital Hamilton population (COHP) analysis as projected from plane-wave basis sets. *J. Phys. Chem. A* **2011**, *115*, 5461–5466.
- (42) Ye, W.; Chen, S.; Lin, Y.; Yang, L.; Chen, S.; Zheng, X.; Qi, Z.; Wang, C.; Long, R.; Chen, M.; Zhu, J.; Gao, P.; Song, L.; Jiang, J.; Xiong, Y. Precisely Tuning the Number of Fe Atoms in Clusters on N-Doped Carbon toward Acidic Oxygen Reduction Reaction. *Chem.* **2019**, *5*, 2865–2878.
- (43) Liang, X.-M.; Wang, H.-J.; Zhang, C.; Zhong, D.-C.; Lu, T.-B. Controlled synthesis of a Ni₂ dual-atom catalyst for synergistic CO₂ electroreduction. *Appl. Catal., B* **2023**, *322*, No. 122073.
- (44) Chen, K.; Wang, J.; Zhang, H.; Ma, D.; Chu, K. Self-Tandem Electrocatalytic NO Reduction to NH₃ on a W Single-Atom Catalyst. *Nano Lett.* **2023**, *23*, 1735–1742.